Olivia Moody

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Education

New York University New York, NY
Bachelor of Arts in Computer Science, Minor in Business | GPA: 3.6 *Jan 2027*
Relevant Coursework: Data Structures & Algorithms, OOP, Operating Systems, Artificial Intelligence, Machine Learning, Computer Architecture, Discrete Mathematics, Data Science II, Statistics with Linear Regression, Linear Algebra, Calculus I

Experience

Software Developer Intern May 2026 – July 2026
Gartner *Dallas, TX*
- Redesigned an internal multi-step Salesforce sales workflow in an Agile Scrum environment using LWC and JavaScript LocalStorage to persist step-level state, allowing high-level sales representatives to edit previous inputs without restarting the workflow and reducing task completion time from 90 minutes to 15–20 minutes.
- Built custom Lightning Web Components featuring asynchronous loading states and visual indicators to prevent UI locking and maintain responsive user feedback during heavy enterprise data retrievals.
- Investigated an undocumented Java backend library for automated PowerPoint generation, uncovering and documenting APIs for speaker note modification before pair-programming a proof-of-concept and engineering a generalized code implementation.

Full Stack Software Engineer Intern June 2025 – Dec 2025
Hopscotch Labs *New York, NY*
- Architected a full-stack user 'mention' system by building Scala and Akka backend services backed by PostgreSQL to parse markdown-formatted mentions and manage data-migration backfill scripts prior to feature rollout.
- Developed custom iOS 'mention' UX using Swift and SwiftUI that implemented @-sign trigger detection and friend-scoped autocomplete, and updated content feed retrieval logic and database schema to dynamically prioritize unread tagged content.
- Integrated server-side version validation endpoints with custom SwiftUI modals to enforce mandatory blocking upgrades and decrease the number of clients on unsupported builds by 90 % .

Server Engineer Intern *Jan 2025 – May 2025*
- Developed an automated keyword vector embedding pipeline in Scala leveraging OpenAI APIs and native cosine similarity matching to canonicalize synonyms and elevate feed personalization.
- Built internal admin dashboards using vanilla JavaScript, HTML, and PostgreSQL to manage keywords, user vectors, and mentions via custom client-side filtering and autocomplete features.

Projects

HedgeLabs | *React, TypeScript, Python, FastAPI, AWS, TimescaleDB, PostgreSQL, Plotly* Oct 2026
- Deployed a full-stack financial risk analytics platform using AWS EC2, RDS, and S3 with TigerData/TimescaleDB to analyze 15+ years of historical high-yield bond ETF data spanning 3,900+ trading days.
- Built interactive Plotly dashboards and FastAPI endpoints for financial simulations and research analytics, integrating time-series market data with Gemini-powered explanations to make complex hedging results interpretable.
- Engineered Python-based backtesting simulations comparing unhedged portfolios, static short positions, and inverse ETF hedges (SJB), modeling $1M portfolios with configurable hedge ratios and quantifying returns, P&L, and drawdowns.
- Conducted quantitative analysis of inverse ETF behavior, measuring -0.970 correlation, -1.005 beta, and 94.17% R^2 across the historical research sample.

D&D Unlimited | *Python, Pycord, Honcho.dev, OpenRouter, GPT-4o, Flowglad, Vercel* Aug 2025
- Shipped the full product within a one-day hackathon, integrating Flowglad payments and a download API so users could purchase and add the bot to any Discord server.
- Placed 3rd of 25 teams at the Plastic Labs x Flowglad Vibe Coding Hackathon by co-building a Discord bot that joins D&D campaigns as AI players modeled on 3 characters from a popular D&D podcast.
- Collaborated on a Python Discord bot that registers a slash command per character and generates in-character responses with GPT-4o via OpenRouter, using Honcho.dev peers and sessions to share campaign memory.
- Wrote a Python script to collect podcast transcripts, sourcing 46K+ words and 4,600+ lines of dialogue across 4 speakers that were parsed into speaker-tagged messages to ground each character's persona.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Scala, C/C++, Swift, SQL, Apex, HTML/CSS
Frameworks & Libraries: React, Node.js, FastAPI, Akka, SwiftUI, Lightning Web Components, pandas, NumPy, Plotly
Developer Tools & Cloud: AWS, PostgreSQL, Docker, Git, Vercel, Cloudflare, Salesforce, Postman, OpenAI API

Interests

Hobbies: Jiu Jitsu, Tetris, Plant Identification, Building Halloween Props, Chinese Rock Paper Scissors, Baking, Pool
Languages: English, Spanish, Mandarin